

Benjamin F. Bubnick

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EDUCATION

- **M.S., Physics** University of Cincinnati, Cincinnati, Ohio December 2010
 - o GPA: 3.3/4.0
- **B.F.A., Fine Art** Ohio University, Athens, Ohio August 2004

PROFESSIONAL EXPERIENCE

Adjunct Professor beginning December 2010
Brown Mackie College, Cincinnati, Ohio

- Lecture in a month-long introductory physics course designed for non-traditional students

Adjunct Professor September 2010 – Present
Miami University, Hamilton, Ohio

- Lecture in introductory physics course designed for non-science majors
- Evaluate and grade students' class work, assignments, and exams
- Provide regularly scheduled office hours to assist students and volunteer extra time outside the office

Teaching Assistant October 2008 – June 2010
University of Cincinnati, Cincinnati, Ohio

- Taught introductory physics labs to 80+ engineers and other non-physics majors yearly in an unsupervised role
- Wrote, administered, and graded quizzes
- Collaborated with colleagues to address teaching issues through regular meetings to establish course objectives
- Taught students how to use Microsoft Excel and DataStudio.
- Taught Mathematica to advanced Aerospace Engineering students

Research Assistant October 2006 – Present
University of Cincinnati, Cincinnati, Ohio

- Determined the physical properties of star clusters to map the structure of our galaxy from photometric data
- Found locations for 100+ stars in each field with 7 sigma precision
- Performed quality control for the Cambridge Astronomical Survey Unit Variable Stars (VVV) database
- Controlled the SpeX spectroscopic tool as a visiting astronomer to the NASA Infrared Telescope Facility

TECHNICAL SKILLS

- Business Software
 - o Adobe Illustrator, Adobe Photoshop, GIMP, Google Docs, Keynote, Microsoft Word, Microsoft PowerPoint, Microsoft Excel, Open Office, Secure Shell Network Protocol, Terminal
- Operating Systems
 - o Linux, MacOS, Unix, Windows
- Programming
 - o IDL, IRAF, HTML, LaTeX, Mathematica, MATLAB, PERL

SCIENTIFIC PUBLICATIONS

1. M. M. Hanson, B. Bubnick
The B Stars of Westerlund 1.
[In preparation].
2. M. M. Hanson, B. Bubnick
Near-Infrared Spectroscopy of Candidates Members of the Galactic Cluster [BDS2003] 107.
Publications of the Astronomical Society of the Pacific, 120, 864.

ACTIVITIES

- American Institute of Aeronautics and Astronautics 2009-Present
 - o Vice President, 2010
 - o Students for the Exploration and Development of Space (SEDS), 2009
- UC Astronomy Club 2008-2010
 - o Secretary, 2009-2010
 - o Treasurer, 2008-2009
 - o Founding Member, 2008
- Society of Physics Students 2005-2008
 - o Treasurer, 2006-2007
 - o Secretary, 2005-2006
- Large Synoptic Survey Telescope (LSST) Seminar Group 2006-2007
- UC-SPARKS: Summer Program for Academic Readiness Knowledge & Success 2006

ACCOMPLISHMENTS

- Mary J Hanna Fellowship Award 2010
- UGS Scholarship for Physics 2008-2010
- Dean's List at UC, GPA 3.4 or higher 2005-2007
- Provost's Undergraduate Research Grant 2003
- Dean's List at OU, GPA 3.5 or higher 2002-2004
- Ohio Proficiency Scholarship 2000

COMMUNITY SERVICE

- Tutor at TCP World Academy 2010-Present
- Project Astro at Ethel M. Taylor Academy 2008
 - o National program to improve the teaching of astronomy and physical science by linking professional and amateur astronomers with local educators
- Sigma Theta Epsilon 2005-2007
 - o Secretary, 2007
 - o Chaplain, 2006
- Reach Out on Campus at Ohio University 2000-2004

UNDERGRADUATE WORK EXPERIENCE

Supplemental Instruction Leader June 2006 – December 2006
University of Cincinnati, Cincinnati, Ohio

- Led group tutoring sessions of up to 12 students in introductory Physics

Student Librarian March 2006 – September 2008
University of Cincinnati, Cincinnati, Ohio

- Provided reference and assisted patrons of the Geology-Math-Physics Library with research needs

Undergraduate Teaching Assistant January 2006 – May 2006
University of Cincinnati, Cincinnati, Ohio

- Assisted the graduate student instructor in the Freshman Engineering Physics recitation course

Productivity Consultant June 2005 – April 2006
FranklinCovey, Cincinnati, Ohio

- Worked to meet customer needs and exceed customer expectations through a strong knowledge of services

Student Researcher October 2003 – May 2004
Ohio University, Athens, Ohio

- Developed a new process for a chemical-etch printing method